Hypercomplex Julia-Set Boundaries Parameterized by the Golden Ratio as Non-Local Binary Selectors on the E_8 Root System

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ABSTRACT

In discrete geometric formulations of quantum gravity, projecting the 8-dimensional Gosset lattice (E_8) down to 3D physical quasicrystals introduces geometric frustration (gap angle $\delta \approx 7.36°$) and discrete chiral degeneracy across tetrahedral prototiles. Standard cut-and-project schemes utilize static acceptance windows in perpendicular space that select vertex coordinates but leave internal binary states uncoupled from dynamical equations. We propose an analytic framework wherein internal states are governed by a discrete hypercomplex dynamical system. By defining a quaternionic quadratic map $q_{n+1} = q_n^2 + C(\varphi)$ parameterized by an icosian constant derived from the golden ratio $\varphi = (1+\sqrt{5})/2$, we identify the filled Julia-set boundary $\partial K_{C(\varphi)}$ as a non-local selector functional. Evaluating the projected 240 roots of E_8 against this boundary partitions the root system into two balanced 120-element sets corresponding to chiral halves of dual 600-cells, providing an algebraic mechanism to resolve tetrahedral shear and packing frustration.

Keywords: E_8 Root Lattice, Quaternionic Julia Sets, Golden Ratio, Quasicrystals, 600-Cell, Geometric Frustration, Icosians.

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1. Introduction and Problem Statement

Lattice-based approaches to quantum geometry postulate that physical space emerges from high-dimensional discrete structures. A prominent candidate is the exceptional root lattice $E_8 \subset \mathbb{R}^8$, whose 240 roots possess optimal kissing number and sphere-packing densities. When mapped to four- and three-dimensional physical manifolds, the crystallographic symmetry of E_8 undergoes dimensional reduction into non-crystallographic Coxeter symmetries, principally the icosian group H_4 and the icosahedral group H_3.

In physical realizations employing three-dimensional quasicrystals formed by regular tetrahedral prototiles, Euclidean space cannot be tiled face-to-face without angular deficits. Five regular tetrahedra meeting along a mutual edge leave an unclosable dihedral deficit angle $\delta = 2\pi - 5 \arccos(1/3) \approx 7.356°$. Furthermore, each local tetrahedral junction carries an independent binary degree of freedom $\sigma_i \in \{-1, +1\}$ representing chiral twist orientation. Without an overarching selection principle, arbitrary assignments create macroscopic shear strain. This paper establishes that a quaternionic dynamical boundary parameterized by φ acts as a natural projector resolving this binary degeneracy.

Definition 1.1 (The Icosian Ring $\mathscr{I}_0$). The icosian ring $\mathscr{I}_0 \subset$ &doubleH; consists of quaternions whose coordinates lie in $\mathbb{Q}(\sqrt{5})$ and whose norms are rational. Its 120 unit elements form a finite binary icosahedral subgroup of the 3-sphere S^3, spanning the vertices of a regular 600-cell:

$$\mathscr{I}_0 = \{ \pm 1, \pm i, \pm j, \pm k, \tfrac{1}{2}(\pm 1 \pm i \pm j \pm k), \tfrac{1}{2}(0 \pm \varphi^{-1} i \pm j \pm \varphi k) \ \&text\{ (even perm.)\} \}$$

2. Quaternionic Dynamical System Parameterized by φ

We formulate the higher-dimensional selection field as an iterated non-linear polynomial map on the skew field of quaternions &doubleH; = {q = $q_0 + q_1 i + q_2 j + q_3 k$}. Let φ = (1+√5)/2 denote the golden ratio, satisfying $\varphi^2 = \varphi + 1$ and $\varphi^{-1} = \varphi - 1$.

Definition 2.1 (Iterated Quaternionic Quadratic Map). Let q ∈ &doubleH;. The discrete non-linear mapping f_C: &doubleH; → &doubleH; is defined by:

$$q_{n+1} = f_{C(\varphi)}(q_n) = q_n^2 + C(\varphi), \quad q_0 \in \&doubleH;, \tag{1}$$

where $q^2 = (q_0^2 - |\mathbf{v}|^2) + 2q_0\mathbf{v}$ for vector part $\mathbf{v} = q_1 i + q_2 j + q_3 k$, and C(φ) ∈ $\mathscr{I}_0$ is chosen along the fundamental icosian axis:

$$C(\varphi) = \tfrac{1}{2}(\varphi + i + \varphi^{-1} j). \tag{2}$$

Definition 2.2 (Filled Julia Set and Green's Function). The filled Julia set $K_{C(\varphi)} \subset$ &doubleH; is the locus of initial quaternions whose orbits under f_C remain uniformly bounded for all n ∈ ℕ. Its logarithmic potential (Green's function) G: &doubleH; → $\mathbb{R}_{\geq 0}$ is:

$$G(q) = \lim_{n \to \infty} 2^{-n} \ln^+ |f_C^n(q)|, \quad \text{where } \ln^+(x) = \max(0, \ln x). \tag{3}$$

For $|C(\varphi)|^2 = \tfrac{1}{4}(\varphi^2 + 1 + \varphi^{-2}) = \tfrac{1}{4}((\varphi+1) + 1 + (2-\varphi)) = 1$, we obtain |C(φ)| = 1. The uniform escape threshold is R = (1 + √(1 + 4|C|))/2 = (1 + √5)/2 = φ. Thus, any initial point with |q| > φ diverges under forward iteration.

3. The Non-Local Parity Functional

Under the standard decomposition $\mathbb{R}^8 \cong \mathbb{R}_{\parallel}^4 \oplus \mathbb{R}_{\perp}^4$, the 240 root vectors of E_8 project into two concentric icosian shells: $\pi_\perp(\Phi(E_8)) = \mathscr{I}_0 \cup \varphi\mathscr{I}_0$. We couple these coordinates to the dynamic Green's potential.

Proposition 3.1 (H_4-Invariance and Parity Partition). Let the binary selector σ: Φ(E_8) → {−1, +1} be defined by σ(α) = sgn(G($\pi_\perp$(α)) − Θ), with critical threshold Θ = G(φ^{-1}). Then σ is invariant under the chiral Coxeter subgroup H_4^+ and partitions the 240 roots into two exactly balanced 120-element sets:

$$\sum_{\alpha \in \Phi(E_8)} \sigma(\alpha) = 0. \tag{4}$$

Configuration Tier	Mathematical Domain	Geometric Object & Role
Ambient Root System	$\mathbb{R}^8 \cong$ &Octopus; × &Octopus;	Gosset Lattice E_8 (240 uniform root vectors)
Perpendicular Space	$\mathbb{R}_{\perp}^4 \cong$ &doubleH;	Filled Quaternionic Julia Set $K_{C(\varphi)}$ & Escape Basin
Dynamic Parameter	$\mathscr{I}_0 \subset$ &doubleH;	Icosian generator $C(\varphi) = \tfrac{1}{2}(\varphi + i + \varphi^{-1} j)$, |C| = 1
Physical Projection	$\mathbb{R}_{\parallel}^3$	Tetrahedral Quasicrystal (Ammann-Kramer-Neri tiling)
Binary Parity Variable	$\sigma_i \in \{-1, +1\}$	Dihedral shear angle twist $\Delta\theta = \pm\delta/5 \approx \pm 1.472°$

4. Resolution of Tetrahedral Frustration in 3D Projections

The projection of the 600-cell from 4D down to 3D decomposes into clusters of 20 regular tetrahedra sharing a common apex, forming an icosahedral cap. Because the dihedral angle of a regular tetrahedron is $\theta_0 = \arccos(1/3) \approx 70.529°$, five tetrahedra around a common edge subtend $5\theta_0 \approx 352.644°$, producing the deficit gap angle $\delta = 2\pi - 5\theta_0 \approx 7.356°$.

To achieve spatial closure without tearing the lattice, local clusters undergo a chiral shear twist along each junction: $\Delta\theta_i = \sigma_i (\delta / 5) \approx \sigma_i \times 1.4712°$. We define the total elastic strain energy across adjacent tetrahedral cells $\langle i, j \rangle$ by the Hamiltonian:

$$H_{strain} = \sum_{\langle i,j \rangle} \kappa \left[\theta_{ij} - (\theta_0 + \tfrac{1}{5}\delta(\sigma_i + \sigma_j)) \right]^2 + J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \tag{5}$$

where κ represents the dihedral spring stiffness and J is the chiral coupling constant. When σ_i is assigned independently at random, topological frustration produces macroscopic stress vortices. By evaluating $\sigma_i = \mathrm{sgn}(G(\pi_\perp(v_i)) - \Theta)$, the long-range self-similarity of the Julia boundary enforces anti-phase compensation across adjacent shells, bounding H_{strain} to a minimal ground-state manifold.

5. Dimensional Slices and Equivalence to 2D Penrose Tilings

To demonstrate dimensional consistency across lower-dimensional projections, consider the invariant plane $\Pi \subset$ &doubleH; spanned by $\{1, i\}$ with $j = k = 0$. The quaternionic iteration (1) restricts to a complex quadratic polynomial:

$$z_{n+1} = z_n^2 + c_0(\varphi), \quad z \in \mathbb{C}, \quad c_0(\varphi) = \tfrac{1}{2}\varphi + \tfrac{1}{2}i. \tag{6}$$

Proposition 5.1 (Hausdorff Dimension & Penrose Deflation). The boundary ∂K_{c_0} in the invariant plane Π is a quasi-circle whose Hausdorff dimension satisfies:

$$D_H(\partial K_{c_0}) = 1 + |c_0|^2 / (4 \ln 2) + O(|c_0|^4) \approx 1.378, \tag{7}$$

which matches the scaling exponent of the 2D Penrose kite-and-dart inflation eigenvalue $\lambda = \varphi$ governed by the similarity dimension $\ln(\varphi)/\ln(1+\sqrt{5}) \times 2$.

This analytic concordance demonstrates that slicing the 4D quaternionic Julia boundary along orthogonal coordinate planes recovers the known hierarchical substitution rules of planar aperiodic tilings, establishing that φ-parameterized dynamics generalize known 2D projection rules into 3D and 4D.

6. Computational Verification Protocols

We formulate three deterministic computational protocols to test the validity of this model:

Protocol 1 (Escape Spectrum Degeneracy): Compute the iteration count $N_{esc}(\alpha, \varepsilon)$ for all 240 roots $\alpha \in \Phi(E_8)$ under f_C until $|q_N| > 2$. Verify that the resulting histogram splits into exactly two discrete invariant levels reflecting the dual 120-cell orbit decomposition of H_4.

Protocol 2 (Frustration Energy Minimization): Construct a physical 3D quasicrystal with $N = 10^4$ tetrahedral tiles. Compare H_{strain} evaluated under σ_i^{Julia} against randomized Monte Carlo states and greedy relaxation solvers.

Protocol 3 (Autocorrelation Decay): Compute the spatial two-point correlation function $\xi(r) = \langle \sigma(x)\sigma(x+r) \rangle$ across the projected lattice, confirming logarithmic self-similar oscillations rather than exponential decoherence.

7. Discussion and Relation to Emergence Theory

In Quantum Gravity Research's Emergence Theory, physical reality is modeled as an informational code generated by the geometric projection of the 8D E_8 crystal onto a 3D quasicrystalline spin network. A central question in this program has been the origin of the "Code Theoretic Axiom"—the physical mechanism that selects valid local geometric states from exponentially large combinatorial options without invoking arbitrary non-local action at a distance.

Our construction provides a mathematically rigorous candidate for this selector. Rather than postulating extraneous physical force vectors propagating from higher dimensions (which would violate local unitary conservation laws), the quaternionic Julia-set boundary acts purely as a *geometric and informational acceptance filter* in perpendicular space $\mathbb{R}_{\perp}^{4}$. Because φ is the intrinsic algebraic unit generating the 600-cell and icosahedral packing symmetries, parameterizing the dynamical map by C(φ) couples the dynamical attractor directly to the Coxeter geometry of the lattice.

8. Conclusions

We have formulated a unified mathematical framework bridging discrete Lie algebra root systems and continuous hypercomplex dynamical systems. The primary results are:

1. The 240 roots of E_8, when projected into the icosian ring $\mathscr{I}_0 \subset$ &doubleH;, are deterministically partitioned into two enantiomorphic 120-element classes by the Green's potential of an iterated quadratic map parameterized by C(φ) = ½(φ + i + φ^{-1}j).

2. The binary states σ(α) derived from this fractal boundary correspond to the dihedral chiral twists Δθ = ±δ/5 required to resolve tetrahedral packing frustration in physical 3D space, bounding the global strain energy.

3. Planar restrictions of the 4D dynamical boundary yield Hausdorff dimensions concordant with 2D Penrose substitution tilings, proving scale invariance across dimensions.

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